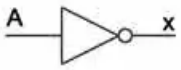
Section 1: Logic Circuits Extension

Xiaofan Chen

x.chen3@massey.ac.nz

Logic Gates

Name	NOT	AND	NAND	OR	NOR	XOR	XNOR
Alg. Expr.	$\overline{A}$	AB	$\overline{AB}$	$A+B$	$\overline{A+B}$	$A \oplus B$	$\overline{A \oplus B}$
Symbol							
Truth Table	A | X 0 | 1 1 | 0	B A | X 0 0 | 0 0 1 | 0 1 0 | 0 1 1 | 1	B A | X 0 0 | 1 0 1 | 1 1 0 | 1 1 1 | 0	B A | X 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 1	B A | X 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 0	B A | X 0 0 | 0 0 1 | 1 1 0 | 1 1 1 | 0	B A | X 0 0 | 1 0 1 | 0 1 0 | 0 1 1 | 1

https://frankcomputerscience.wordpress.com/chapter-3/

Boolean Algebra

The Axioms of Boolean Algebra

P + Q = Q + P
P ▪ Q = Q ▪ P

(P + Q) + R = P + (Q + R)
(P ▪ Q) ▪ R = P ▪ (Q ▪ R)

P ▪ Q + P ▪ R = P ▪ (Q + R)
(P + Q) ▪ (P + R) = P + (Q ▪ R)

TRUE + P = TRUE
FALSE + P = P
TRUE ▪ P = P
FALSE ▪ P = FALSE

P + ~P = TRUE
P ▪ ~P = FALSE
P + P = P
P ▪ P = P
~~P = P

P + ~P ▪ Q = P + Q
P + P ▪ Q = P

De Morgan's Laws

~(P ▪ Q) = ~P + ~Q
~(P + Q) = ~P ▪ ~Q

Boolean Expression 1

$$(A+B)\ (\sim A + C)\ (B+C)$$

$$= (A\sim A + AC + \sim AB + BC)\ (B+C)$$

$$= (AC + \sim AB + BC)\ (B+C)$$

$$= ABC + ACC + \sim ABB + \sim ABC + BBC + BCC$$

$$= ABC + AC + \sim AB + \sim ABC + BC + BC$$

$$= ABC + AC + \sim AB + \sim ABC + BC$$

$$= AC(B+1) + \sim AB\ (1+C) + BC$$

$$= AC + \sim AB + BC$$

$$(A+B)\ (\sim A + C)\ (B+C)$$

$$= (A+ B)\ (\sim AB+C)$$

$$= A\sim AB + AC + \sim ABB + BC$$

$$= 0 + AC + \sim AB + BC$$

$$= AC + \sim AB + BC$$

De Morgan's Laws

$$\sim(P \cdot Q) = \sim P + \sim Q$$

$$\sim(P + Q) = \sim P \cdot \sim Q$$

The Axioms of Boolean Algebra

$$P + Q = Q + P$$

$$P \cdot Q = Q \cdot P$$

$$(P + Q) + R = P + (Q + R)$$

$$(P \cdot Q) \cdot R = P \cdot (Q \cdot R)$$

$$P \cdot Q + P \cdot R = P \cdot (Q + R)$$

$$(P + Q) \cdot (P + R) = P + (Q \cdot R)$$

$$TRUE + P = TRUE$$

$$FALSE + P = P$$

$$TRUE \cdot P = P$$

$$FALSE \cdot P = FALSE$$

$$P + \sim P = TRUE$$

$$P \cdot \sim P = FALSE$$

$$P + P = P$$

$$P \cdot P = P$$

$$\sim\sim P = P$$

$$P + \sim P \cdot Q = P + Q$$

$$P + P \cdot Q = P$$

Boolean Expression 2

$$(A+B)\ (\sim A + C)\ (B+C)$$

$$= (A+ B)\ (\sim AB+C)$$

$$= A\sim AB + AC + \sim ABB + BC$$

$$= 0 + AC + \sim AB + BC$$

$$= AC + \sim AB + BC$$

Apply Consensus/Redundancy theorem

$$(A+B)\ (\sim A + C)\ (B+C)$$

$$= AC + \sim AB + BC$$

$$= AC + \sim AB + BC\ (A + \sim A)$$

$$= AC + \sim AB + ABC + \sim ABC$$

$$= AC\ (1 + B) + \sim AB\ (1 + C)$$

$$= AC + \sim AB$$

De Morgan's Laws

$$\sim(P \cdot Q) = \sim P + \sim Q$$

$$\sim(P + Q) = \sim P \cdot \sim Q$$

The Axioms of Boolean Algebra

$$P + Q = Q + P$$

$$P \cdot Q = Q \cdot P$$

$$(P + Q) + R = P + (Q + R)$$

$$(P \cdot Q) \cdot R = P \cdot (Q \cdot R)$$

$$P \cdot Q + P \cdot R = P \cdot (Q + R)$$

$$(P + Q) \cdot (P + R) = P + (Q \cdot R)$$

$$TRUE + P = TRUE$$

$$FALSE + P = P$$

$$TRUE \cdot P = P$$

$$FALSE \cdot P = FALSE$$

$$P + \sim P = TRUE$$

$$P \cdot \sim P = FALSE$$

$$P + P = P$$

$$P \cdot P = P$$

$$\sim\sim P = P$$

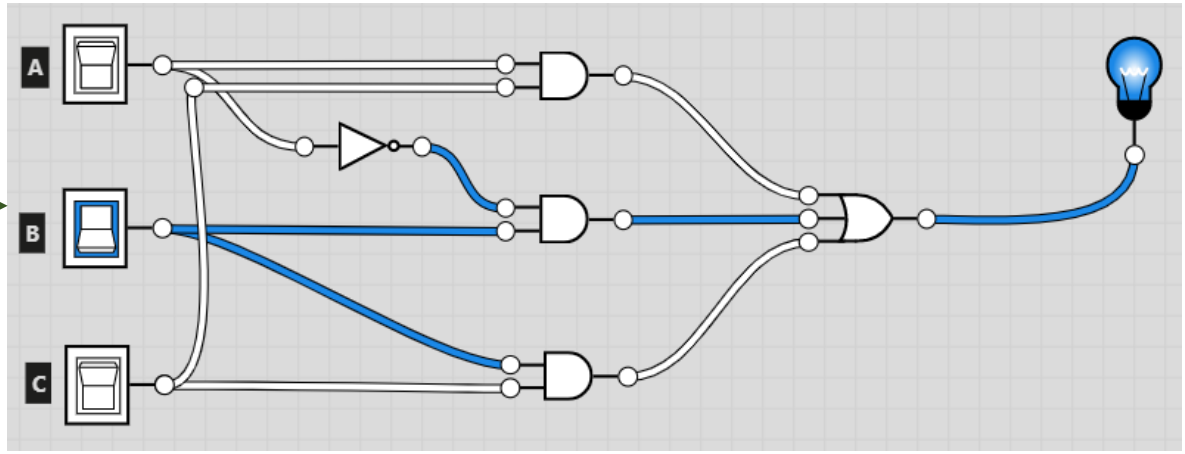
$$P + \sim P \cdot Q = P + Q$$

$$P + P \cdot Q = P$$

Boolean Expression → Logic Circuit

(A+B) (~A + C) (B+C)

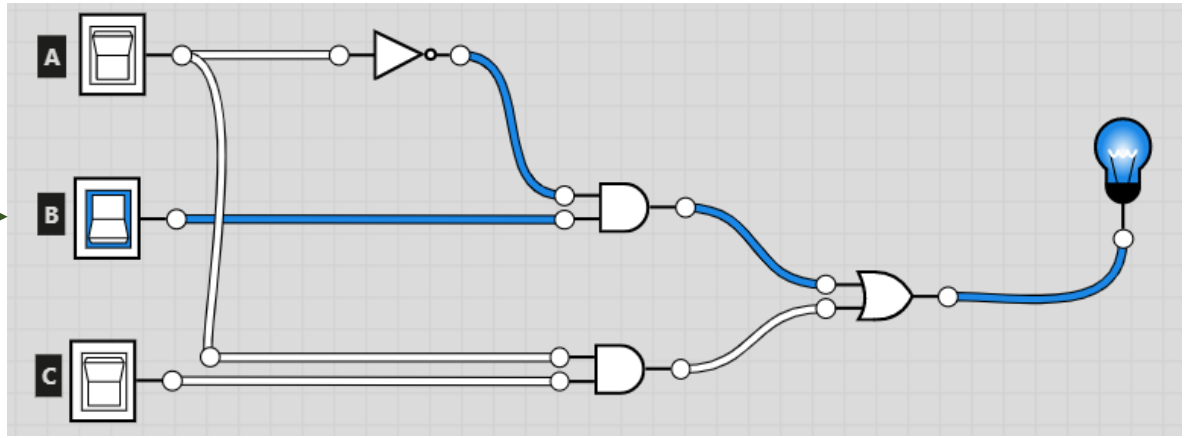
= AC + ~AB + BC



(A+B) (~A + C) (B+C)

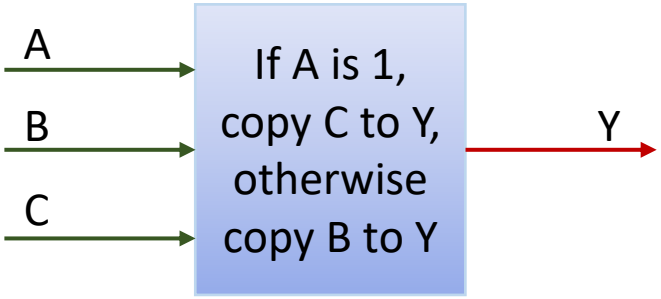
= AC + ~AB + BC

= AC + ~AB



Design a Logic Circuit

- Understand the problem
- Draw a truth table
- Create a Boolean expression
- Reduce Boolean expression
- Design a logic circuit
- Check that circuit matches the truth table



A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$\sim AB\sim C + \sim ABC + A\sim BC + ABC$$
$$= \sim AB\,(\sim C + C) + AC\,(\sim B + B)$$
$$= \sim AB + AC$$

